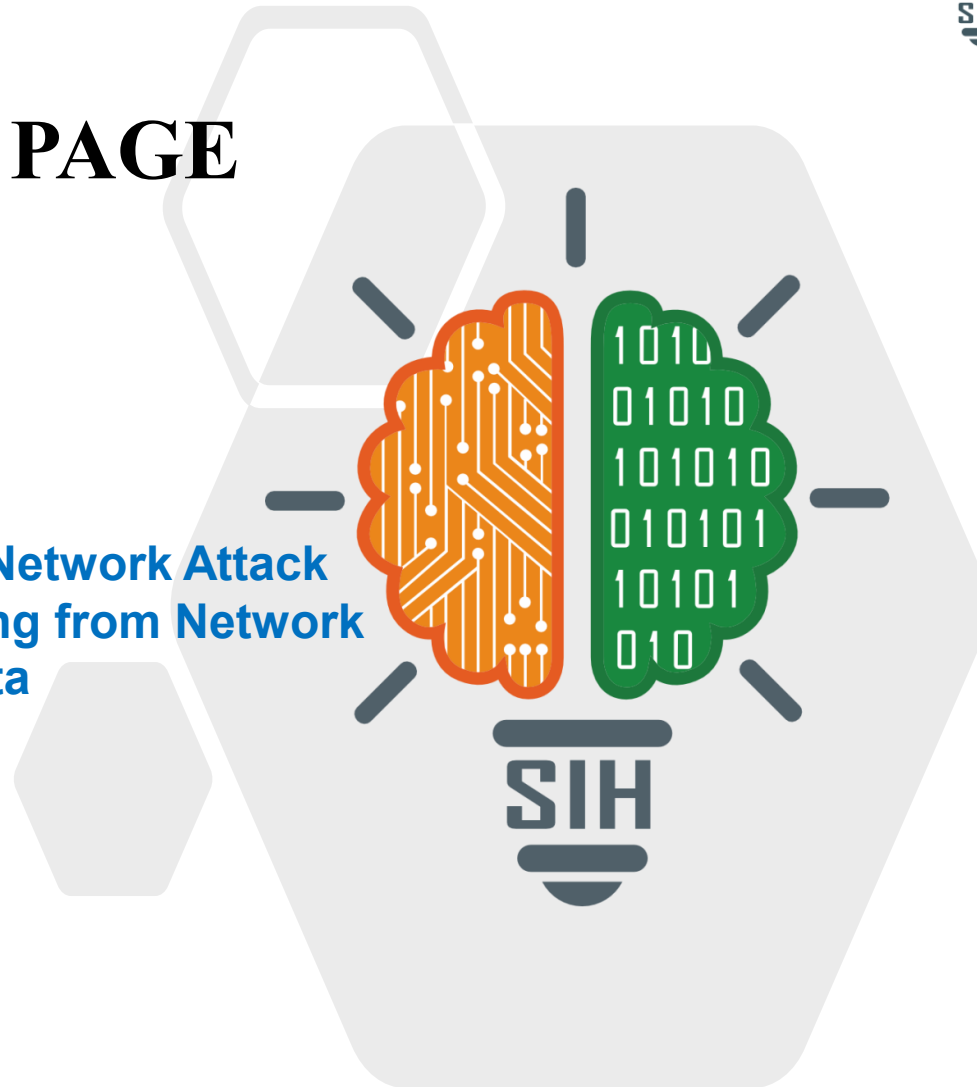


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – **SIH26153**
- Problem Statement Title- **AI based Network Attack Forecasting from Network Traffic Data**
- Theme- **Blockchain & Cybersecurity**
- PS Category- **Software**
- Team ID- **149432**
- Team Name- **GNC NIVARAN**



TO TRANSFORM CYBERSECURITY FROM REACTIVE ATTACK DETECTION INTO PROACTIVE ATTACK FORECASTING.

PROBLEM OVERVIEW

1. Reactive Detection

Traditional IDS detects attacks after suspicious activity occurs.

02. Missing Attack Progression

Attacks evolve over time, but individual flows may hide the sequence.

03. No Future Visibility

Defenders lack insight into the attacker’s likely next move.

THE GAP:

Detect the present → Cannot forecast the next stage

CORE PROBLEM

How can we learn the evolving state of a network and forecast the future progression of a cyber attack before compromise?

PROPOSED SOLUTION

01. COLLECT NETWORK TRAFFIC

PCAP / Flow Records → Packets, Ports, TCP Flags, Timing, Bytes

↓

02. BUILD NETWORK STATES

Feature Extraction → Time-Windowed Traffic States

↓

03. LEARN TEMPORAL BEHAVIOUR

World Model → Learns how network states evolve over time

↓

04. FORECAST FUTURE STATES

LSTM / Transformer → K-Step Future-State & Infiltration Prediction

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05. ANALYZE ATTACK TRAJECTORY

Predicted behaviour → Attack Stages + MITRE ATT&CK

↓

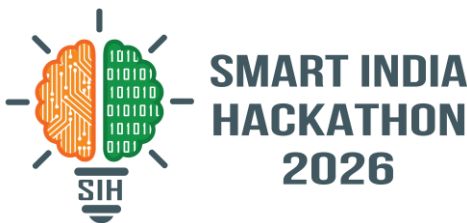
06. EXPLAIN & ALERT

SHAP / Attention → Risk + Confidence → Defender Dashboard

CORE SOLUTION

Traffic → Network State → Learn Temporal Behaviour → Forecast
Attack → Explain → Defend

TECHNICAL APPROACH

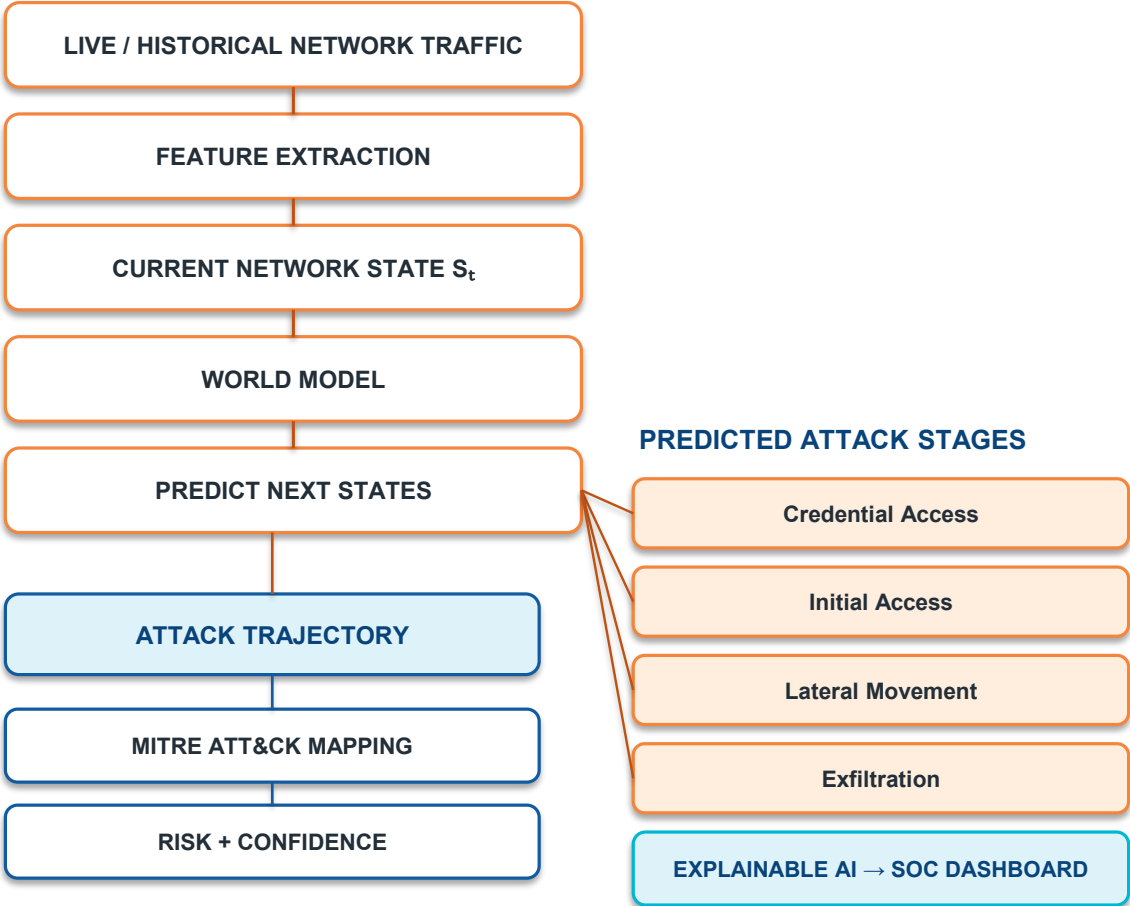


METHODOLOGY



TRAFFIC → STATE → FORECAST → EXPLAIN → DEFEND

FLOW CHART



TECH STACK



FEASIBILITY AND VIABILITY



ANALYSIS OF THE FEASIBILITY OF THE IDEA

1. TECHNICAL FEASIBILITY Built with mature, open-source technologies: Python, PyTorch, FastAPI and React. Public PCAP/flow datasets and replayable traffic make the forecasting prototype practical without special hardware.	2. DATA & MODEL FEASIBILITY CIC-IDS-2018 / CTU-13 and related telemetry can support experimentation. Temporal windowing converts traffic into network-state sequences for LSTM, GRU and Transformer comparisons.	3. OPERATIONAL FEASIBILITY A modular pipeline—traffic ingestion, state builder, forecasting engine and SOC dashboard—allows staged development and easy integration with existing monitoring workflows.	4. SECURITY & DEPLOYMENT Offline/local processing can keep sensitive telemetry within the controlled environment. Human-in-the-loop recommendations avoid unsafe automatic blocking during the prototype stage.
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POTENTIAL CHALLENGES AND RISKS

STRATEGIES FOR OVERCOMING THESE CHALLENGES

SEQUENTIAL DATA LIMITATION

Most intrusion datasets are designed for attack classification, not next-stage forecasting.

CLASS IMBALANCE & FALSE ALERTS

Rare attack stages and noisy telemetry can bias models or produce overconfident forecasts.

FORECAST DEGRADATION

Accuracy can decrease as the model predicts multiple future states farther ahead.

REAL-TIME SCALABILITY

High-volume packet/flow streams may increase preprocessing and inference latency.

TEMPORAL SEQUENCE CONSTRUCTION

Preserve timestamps, hosts, sessions and attack scenarios; build time-windowed state-transition labels instead of randomly shuffling rows.

UNCERTAINTY-AWARE FORECASTING

Show probabilities and confidence for each predicted stage; treat trajectories as possibilities, not guaranteed outcomes.

ROBUST MODEL EVALUATION

Use class-aware training, calibrated probabilities, confidence thresholds, F1/FPR and forecasting-specific metrics.

MODULAR & EFFICIENT PIPELINE

Aggregate traffic into flow/state windows, optimize feature processing and keep AI, API and dashboard components independently scalable.

IMPACT AND BENEFITS

UNIQUE FEATURES

ATTACK-STAGE FORECASTING

Predicts probable next attacker stage instead of only classifying current traffic.

MULTI-STEP TRAJECTORY

Forecasts how an attack may progress across multiple future network states.

MITRE ATT&CK CONTEXT

Maps supported observed and predicted behaviour into analyst-readable ATT&CK context.

EXPLAINABLE FORECASTS

Shows the traffic features and temporal signals that contributed to a prediction.

EARLY-WARNING INTELLIGENCE

Combines probability, confidence, risk and defensive recommendations in a SOC view.

IMPACTS

PROACTIVE CYBER DEFENCE

Shifts security from only reacting to detected events toward anticipating probable attacker progression.

FASTER SOC RESPONSE

Earlier warning gives analysts more time to investigate, prioritize, isolate and contain threats.

CRITICAL INFRA READINESS

Supports forward-looking monitoring for enterprise, government and critical-information networks.

BETTER ANALYST DECISIONS

MITRE context, confidence and explanations make forecasts easier to interpret and validate.

BENEFITS

EARLY WARNING TIME

Measures how much lead time the forecast provides before the next attack stage occurs.

REDUCED ALERT OVERLOAD

Prioritizes likely progression and risk instead of presenting every suspicious event equally.

EXPLAINABLE & TRUSTWORTHY

Feature attribution helps analysts verify why the model expects a particular progression.

SCALABLE & MODULAR

Independent ingestion, AI, API and dashboard layers support gradual enterprise-scale expansion.

CORE VALUE: Detect what is happening → Forecast what could happen next → Give defenders a head start

RESEARCH AND REFERENCES



DETAILS / LINKS OF REFERENCE & RESEARCH WORK

MITRE ATT&CK®	Adversary tactics & techniques knowledge base used to contextualize observed and forecast behaviour. attack.mitre.org
CSE-CIC-IDS2018	Realistic network traffic and attack scenarios with flow features; useful for traffic modelling and prototype experiments. UNB / Canadian Institute for Cybersecurity
CTU-13 Dataset	Thirteen botnet scenarios with malicious, normal and background traffic; useful for temporal scenario analysis. Stratosphere Laboratory, CTU
Model & XAI Research	Sequence models (LSTM / GRU / Transformer), probability calibration and SHAP-based feature attribution form the experimental research direction. Temporal forecasting + explainability literature

OFFICIAL LINKS

[MITRE ATT&CK • attack.mitre.org](#)

[CSE-CIC-IDS2018 • unb.ca/cic/datasets](#)

[SHAP • shap/researchpaper.io](#)

[CTU-13 • stratosphereips.org/datasets](#)

[IEEE XPLORE • ieee.xplore](#)

Research gap: preserve temporal order and construct network-state sequences for next-stage / multi-step forecasting.

PIPELINE

